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Statistics

Mean questions are total questions: $\text{mean} \times \text{number} = \text{total}$. Grouped tables use midpoints. Compare with an average and a spread.

THE TOPIC ON ONE MAP

Statistics

01 – Averages from a list

mean, median, mode, range missing numbers

02 – Mean problems

$\text{total} = \text{mean} \times n$ add or remove one

03 – Grouped tables

modal class midpoints estimate

04 – Quartiles and IQR

Q_1, Q_3 compare two sets

05 – Charts

pie bar time series

FOUR MEASURES Three averages and one spread:

ZONE MEAN

$\text{total} \div \text{how many}$
uses every value

ZONE MEDIAN

the middle
position $\frac{n+1}{2}$

ZONE MODE

most often
grouped: modal
class

ZONE RANGE / IQR

the spread
big – small;
 $Q_3 - Q_1$

01 Averages and Range from a List

Order the list first. For puzzle questions, draw the boxes in order and fill in what each fact forces: the mode repeats, the median sits in the middle, the range fixes the ends.

WHEN YOU SEE → YOU DO

“the median is 7” with five numbers

→ the 3rd box is 7

“the median is 8.5” with an even count

→ the two middle boxes add to 17

“the mode is 8”

→ 8 appears most — at least twice

“the range is 5”

→ largest – smallest = 5

“which average is appropriate?”

→ an extreme value → median, not mean

WORKED MODEL • JUNE 2022 P2H

$a < b < c < d$; the list a, b, c, d, d, d has mode 9, median 8, range 4.

1 mode: $d = 9$

repeated

2 median: $\frac{c+9}{2} = 8 \Rightarrow c = 7$

3rd and 4th

3 range: $9 - a = 4 \Rightarrow a = 5$, then $b = 6$

ends

ANSWER $a = 5, b = 6, c = 7, d = 9$

IDEA 01

Practise: Averages and Range from a List

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Averages

CH 6 • TOPIC 1 3 marks

Find the mean, median, mode and range of 4, 9, 2, 9, 6, 3, 9, 6.

Q2 DR ESLAM Missing numbers

CH 6 • TOPIC 1 3 marks

Four whole numbers have mode 5, median 6 and mean 7. Find the four numbers.

SELECTED PRACTICE

Q3**PRACTICE**

Practice • Easy

CH 6 • TOPIC 1 3 marks

5 The students in Class A and in Class B take the same examination.

The lowest score in Class A is 39

The range of scores for Class A is 57

The lowest score in Class B is 33

The range of scores for Class B is 60

Find the range of scores for all the students in both classes.

(3 marks)**Q4****PRACTICE**

Practice • Easy

CH 6 • TOPIC 1 3 marks

12 Six positive numbers have

a mean of 10

a range of 19

Four of the numbers are 12 7 15 3

Work out the other two numbers.

(3 marks)

Q5**PRACTICE**

Practice • Easy

CH 6 • TOPIC 1**4 marks**

- 18** Jenny played four games of golf.
For these games her modal score was 76 and her mean score was 75.
Her range of scores was 10.

What were her scores for the four games?

(4 marks)

Q6

PRACTICE Practice • Medium

CH 6 • TOPIC 1 3 marks

PRACTISE • 6.1

14 Here is some information about 26 houses.

a , b and c are all **different** numbers.

Number of bedrooms	Number of houses
1	7
2	a
3	b
4	c
5	8

The median number of bedrooms is 3.5

Work out a possible set of values for a , b and c .

$a =$
 $b =$
 $c =$
(3 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q4

CH 6 • TOPIC 1 4 marks

- 4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q2

CH 6 • TOPIC 1 • 4 marks

2 Given that $a < b < c$

the four whole numbers a , a , b and c have

a mode of 7

a median of 8.5

a mean of 9

Work out the value of a , the value of b and the value of c .

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 2 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q2

CH 6 • TOPIC 1 3 marks

- 2 Ava writes down five whole numbers.

For these five numbers

the median is 7

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

.....
(Total for Question 2 is 3 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q5

CH 6 • TOPIC 1**4 marks**

5 Yusuf sat 8 examinations.

Here are his marks for 5 of the examinations.

68 72 75 77 80

For his results in all 8 examinations

the mode of his marks is 80

the median of his marks is 74

the range of his marks is 16

Find Yusuf's marks for each of the other 3 examinations.

.....
.....
.....

(Total for Question 5 is 4 marks)

Q11 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 1H • Q5**CH 6 • TOPIC 1** **6 marks**

5 Jenny has six cards.

Each card has a whole number written on it so that

- the smallest number is 5
- the largest number is 24
- the median of the six numbers is 14
- the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

(a) For the remaining four cards, write on each dotted line a number that could be on the card.

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

(b) Work out the number of points the team scored in its 6th game.

.....
(Total for Question 5 is 6 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q3

CH 6 • TOPIC 1**3 marks**

- 3 Alberto, Bill, Candela and Diana are four friends.

Here is some information about the height of each of these friends.

Alberto's height is 158 cm.

Bill's height is 175 cm.

Candela's height is greater than Diana's height.

The median height of these four friends is 160 cm.

The range of the heights of these four friends is 21 cm.

Work out Candela's height and Diana's height.

Candela cm

Diana cm

(Total for Question 3 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q1

CH 6 • TOPIC 1 3 marks

You must write down all the stages in your working.

- 1 Here are some integers where $a < b < c < d$

$a \qquad b \qquad c \qquad d \qquad d \qquad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 1 is 3 marks)

Q14 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2HR • Q3**CH 6 • TOPIC 1** **2 marks**

- 3 Here is a list of four numbers written in ascending order of size

x x y 15

where x and y are integers.

The numbers have

a median of 12.5

a range of 4

Find the value of x and the value of y

$x =$

$y =$

(Total for Question 3 is 2 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q1

CH 6 • TOPIC 1**3 marks**

You must write down all the stages in your working.

- 1** Here are eight numbers written in order of size

h 6 7 8 j 16 k k

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

$h = \dots\dots\dots$

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 1 is 3 marks)

Q16 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2H • Q1**CH 6 • TOPIC 1** **3 marks**

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q4

CH 6 • TOPIC 1 3 marks

4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

$e =$

$f =$

$g =$

(Total for Question 4 is 3 marks)

02 Mean Problems: Work with Totals

Never average the averages. Turn every mean into a total (mean $\times$ number), add or subtract totals, then divide again.

WHEN YOU SEE $\rightarrow$ YOU DO

“the mean of 5 is 28; 2 get off”

$\rightarrow$ total before – total removed

“one more is added, the new mean is ...”

$\rightarrow$ new total – old total = the new value

two groups with different means

$\rightarrow$ add the two totals, divide by everyone

“at least 77 for all 6”

$\rightarrow$ need total $\geq 6 \times 77$

Mean $\times$ number = total

Add or subtract totals

Divide by the new number

WORKED MODEL • JANUARY 2020 P1HR

The mean of 5 children is 28 kg. Two with mean 26.5 kg get off. Mean of the other 3?

1 $5 \times 28 = 140, 2 \times 26.5 = 53$

totals

2 $140 - 53 = 87$

remaining total

ANSWER $87 \div 3 = 29 \text{ kg}$

× **TRAP** $\frac{70+80}{2} = 75$ is wrong when the groups are different sizes. Use the totals.

IDEA 02

Practise: Mean Problems: Work with Totals

My questions first, then selected practice, then real past papers.

Q18 **DR ESLAM** Totals

CH 6 • TOPIC 1 3 marks

The mean of 6 numbers is 15. One number is removed and the mean becomes 14. Which number was removed?

SELECTED PRACTICE**Q19** **PRACTICE** Practice • Easy

CH 6 • TOPIC 1 3 marks

14 The mean mass of a squad of 19 hockey players is 82 kg

A player of mass 93 kg joins the squad.

Work out the mean mass of the squad now.

.....kg

(3 marks)

Q20**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1**3 marks**

- 1 Hertford Juniors is a basketball team.

At the end of 10 games, their mean score is 35 points per game. At the end of 11 games, their mean score has gone down to 33 points per game.

How many points did the team score in the 11th game?

(3 marks)

Q21**PRACTICE**

Practice · Hard

CH 6 • TOPIC 1 **3 marks**

5 Walkden Reds is a basketball team.

At the end of 11 games, their mean score was 33 points per game.

At the end of 10 games, their mean score was 2 points higher.

Jordan says,

"Walkden Reds must have scored 13 points in their 11th game."

Is Jordan right?

You must show how you get your answer.

(3 marks)

Q22**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1 4 marks

16 Here are two sets of numbers, A and B.

Set A

200	160
104	100

Set B

270	400	483
300	x	

mean of Set A : mean of Set B = 3 : 8

Work out the value of x .

(4 marks)

PAST PAPERS

Q23 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 1HR • Q8**CH 6 • TOPIC 1** **3 marks**

- 8 5 children are playing on a trampoline.
The mean weight of the 5 children is 28 kg.

2 of the children get off the trampoline.
The mean weight of these 2 children is 26.5 kg.

Work out the mean weight of the 3 children who remain on the trampoline.

..... kg

(Total for Question 8 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 2H • Q7**CH 6 • TOPIC 1** **3 marks**

- 7 There are 5 cocoa pods in a bag.
The mean weight of the 5 cocoa pods is 398 grams.

A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.

Work out the weight of the sixth cocoa pod that is put into the bag.

..... grams

(Total for Question 7 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q10

CH 6 • TOPIC 1 3 marks

10 Jethro has sat 5 tests.

Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

(Total for Question 10 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q8

CH 6 • TOPIC 1 3 marks

- 8 A scientist is investigating the weight of 50 tigers.

Here is some information about these tigers.

	Type of tiger	
	Siberian	Bengal
Number of tigers	22	28
Mean weight of tigers (kg)	260	

The mean weight of all 50 tigers is 218 kg

Work out the mean weight of the Bengal tigers.

..... kg

(Total for Question 8 is 3 marks)

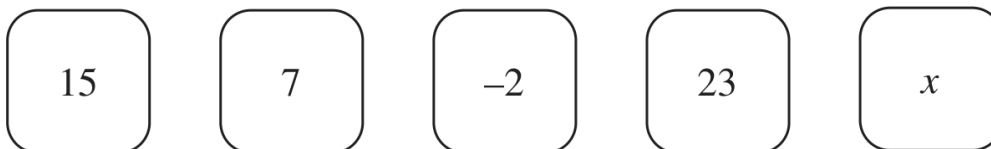
Q27**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q3

CH 6 • TOPIC 1 3 marks

3 Here are five cards.

Each card has a number written on it.



The mean of the five numbers is 12

Work out the value of x

$x =$

(Total for Question 3 is 3 marks)

Q28 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2H • Q12**CH 6 • TOPIC 1** **3 marks**

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

$W = \dots\dots\dots$

(Total for Question 12 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q10

CH 6 • TOPIC 1 3 marks

10 Gemara works as a taxi driver.

Last week, he recorded the following information about the distances he drove.

For the 5 days from Monday to Friday, the mean number of kilometres he drove was 104

For the 7 days from Monday to Sunday, the mean number of kilometres he drove was 127

On Saturday, Gemara drove 132 kilometres.

Work out the number of kilometres he drove on Sunday.

..... kilometres

(Total for Question 10 is 3 marks)

Q30 **PAST PAPER** Edexcel 4MA1 • June 2023 • Paper 2H • Q10**CH 6 • TOPIC 1** **4 marks**

10 Team A and Team B take part in a quiz league.

After 11 rounds, Team A has a mean score per round of 17

After 9 rounds, Team B has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team A scored more points than Team B.

How many more?

(Total for Question 10 is 4 marks)

Q31 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 2H • Q8**CH 6 • TOPIC 1** **3 marks**

8 The mean number of goals scored by a hockey team in 8 matches is 6

The team plays 2 more matches and scores k goals in each match.

The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

$k =$

(Total for Question 8 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q12

CH 6 • TOPIC 1**3 marks**

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800

For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

.....
(Total for Question 12 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q1

CH 6 • TOPIC 1 3 marks**You must write down all the stages in your working.****1** Here are six cards.

Five of the cards have a number written on them.

16	15	3	2	9	
----	----	---	---	---	--

Work out the number that should be written on the last card so that the mean of the six numbers will be 11

(Total for Question 1 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2024 • Paper 2H • Q8**CH 6 • TOPIC 1** **3 marks**

- 8 Kim puts 4 bananas in a bag.
The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.
The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

(Total for Question 8 is 3 marks)

Q35 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2HR • Q8**CH 6 • TOPIC 1** **3 marks**

- 8 The mean of 7 numbers is 60

The mean of 3 of the numbers is 46

Work out the mean of the other 4 numbers.

.....

(Total for Question 8 is 3 marks)

03 Grouped Frequency Tables

You do not know the exact values, so use the midpoint of each class. Estimate of total = $\sum fx$; estimate of mean = $\sum fx \div \sum f$.

WHEN YOU SEE → YOU DO

“write down the modal class”

→ the class with the biggest frequency

“estimate the mean”

→ midpoint $\times$ frequency, add, $\div$ total frequency

“estimate the total”

→ $\sum fx$ only — do not divide

“class containing the median”

→ position $\frac{n}{2}$ in the running total

the mean is given, find x

→ form an equation with $\sum fx$

class	f	midpoint x	$f \times x$
$40 < s \leq 50$	10	45	450
$50 < s \leq 60$	16	55	880
...	...	...	...
total	80		5310

$$\text{mean} \approx 5310 \div 80 = 66.4$$

× **TRAP** Dividing by the number of classes (5) instead of the total frequency (80).

IDEA 03

Practise: Grouped Frequency Tables

My questions first, then selected practice, then real past papers.

Q36**DR ESLAM**

Grouped mean

CH 6 • TOPIC 1

4 marks

Estimate the mean: $0 < t \leq 10$: 4, $10 < t \leq 20$: 10, $20 < t \leq 30$: 6.

SELECTED PRACTICE

Q37**PRACTICE**

Practice • Easy

CH 6 • TOPIC 1 1 mark

- 4 The table shows information about the weights, in kg, of 40 parcels.

Weight of parcel (p kg)	Frequency
$0 < p \leq 1$	19
$1 < p \leq 2$	12
$2 < p \leq 3$	5
$3 < p \leq 4$	2
$4 < p \leq 5$	2

Write down the modal class.

(1 mark)

Q38**PRACTICE**

Practice • Medium

CH 6 • TOPIC 1**3 marks**

15 200 people recorded the time they spent on social media one day.

The table shows the results.

Time, t (mins)	Frequency	Midpoint	
$0 \leq t < 30$	24		
$30 \leq t < 50$	76		
$50 \leq t < 60$	52		
$60 \leq t < 90$	48		
	Total = 200		

Work out an estimate of the mean time.

.....mins

(3 marks)

Q39**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1 4 marks

- 8 Sumeet records the times, in minutes, for 40 runners to finish a half marathon. Information about these times is shown in the table.

Time (t minutes)	Frequency
$60 < t \leq 90$	10
$90 < t \leq 120$	14
$120 < t \leq 150$	9
$150 < t \leq 180$	5
$180 < t \leq 210$	2

Calculate an estimate for the mean time.

(4 marks)

Q40**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1

5 marks**13 (a)** Jenny works in a shop that sells belts.

The table shows information about the waist sizes of 50 customers who bought belts from the shop in May.

Belt size	Waist(w inches)	Frequency
Small	$28 < w \leq 32$	24
Medium	$32 < w \leq 36$	12
Large	$36 < w \leq 40$	8
Extra Large	$40 < w \leq 44$	6

Calculate an estimate for the mean waist size.

(3 marks)**(b)** Belts are made in sizes Small, Medium, Large and Extra Large.

Jenny needs to order more belts in June.

The modal size of belts sold is Small.

Jenny is going to order $\frac{3}{4}$ of the belts in size Small. The manager of the shop tells Jenny she should **not** order so many Small belts.

Who is correct, Jenny or the manager?

You must give a reason for your answer.

(2 marks)

PAST PAPERS

Q41 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 1H • Q4

CH 6 • TOPIC 1 7 marks

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

.....
- (b) Work out an estimate for the mean weight of the 40 babies.

..... kg

One of the 40 babies is going to be chosen at random.

- (c) Find the probability that this baby has a weight of more than 5 kg.

.....

(Total for Question 4 is 7 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q2

CH 6 • TOPIC 1 3 marks

- 2 The table gives information about the amount of money, in £, that Fiona spent in a grocery store each week during 2019

Amount spent (£ x)	Frequency
$0 \leq x < 20$	5
$20 \leq x < 40$	11
$40 \leq x < 60$	8
$60 \leq x < 80$	19
$80 \leq x < 100$	9

Work out an estimate for the total amount of money that Fiona spent in the grocery store during 2019

£.....

(Total for Question 2 is 3 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q4

CH 6 • TOPIC 1 5 marks

- 4 The table gives information about the length of time, in minutes, that each of 60 students took to travel to school on Monday.

Length of time (t minutes)	Frequency
$0 < t \leq 10$	4
$10 < t \leq 20$	10
$20 < t \leq 30$	15
$30 < t \leq 40$	25
$40 < t \leq 50$	6

- (a) Write down the modal class interval.

.....

- (b) Work out an estimate for the mean length of time taken by these 60 students to travel to school on Monday.

Give your answer correct to one decimal place.

..... minutes

(Total for Question 4 is 5 marks)

Q44

PAST PAPER

Edexcel 4MA1 • January 2021 • Paper 1H • Q2

CH 6 • TOPIC 15 marks

PRACTISE • 6.1

- 2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	Frequency
$40 < s \leq 50$	10
$50 < s \leq 60$	16
$60 < s \leq 70$	19
$70 < s \leq 80$	23
$80 < s \leq 90$	12

- (a) Write down the modal class.
-
- (b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge.
Give your answer correct to 3 significant figures.

..... kilometres per hour

(Total for Question 2 is 5 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q10

CH 6 • TOPIC 1 4 marks

- 10** A mathematics teacher at a school asked a group of students how far, in kilometres, each student had travelled to get to school that day.

The table gives information about their answers.

Distance travelled (d km)	Number of students
$0 < d \leq 2$	x
$2 < d \leq 4$	11
$4 < d \leq 6$	8
$6 < d \leq 8$	6
$8 < d \leq 10$	5

The teacher calculated that an estimate for the mean distance travelled by the whole group of students was 4.25 km.

Work out the value of x .

Show your working clearly.

$x =$

(Total for Question 10 is 4 marks)

Q46**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q3

CH 6 • TOPIC 1**5 marks**

- 3 The table gives information about the amounts of money, in euros, that 70 of Anjali's friends spent last Saturday.

Money spent (S euros)	Frequency
$0 < S \leq 8$	6
$8 < S \leq 16$	14
$16 < S \leq 24$	19
$24 < S \leq 32$	25
$32 < S \leq 40$	6

One of Anjali's 70 friends is going to be chosen at random.

- (a) Find the probability that this friend spent more than 24 euros last Saturday.

- (b) Work out an estimate for the mean amount of money spent by Anjali's friends last Saturday.
Give your answer correct to 2 decimal places.

..... euros

(Total for Question 3 is 5 marks)

Q47**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q9

CH 6 • TOPIC 1 4 marks

- 9 The frequency table gives information about the number of points scored by a player.

Number of points	Frequency
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$

(Total for Question 9 is 4 marks)

Q48**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q2

CH 6 • TOPIC 1 3 marks

- 2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leq 30$	5
$30 < M \leq 60$	6
$60 < M \leq 90$	8
$90 < M \leq 120$	9
$120 < M \leq 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

..... minutes

(Total for Question 2 is 3 marks)

You must write down all the stages in your working.

1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (<i>m</i> minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

(a) Write down the modal class.

(b) Work out an estimate for the total length, in minutes, of these telephone calls.

..... minutes

(Total for Question 1 is 4 marks)

You must write down all the stages in your working.

1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

(a) Write down the modal class.

.....

(b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars

(Total for Question 1 is 5 marks)

PRACTISE • 6.1

You must write down all the stages in your working.

1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

(a) Write down the modal class.

.....

(b) Work out an estimate for the mean distance.

..... km

(Total for Question 1 is 5 marks)

PRACTISE • 6.1

Q52

PAST PAPER

Edexcel 4MA1 • November 2025 • Paper 1H • Q1

CH 6 • TOPIC 14 marks

PRACTISE • 6.1

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

04 Quartiles, IQR and Comparing

For n ordered values: Q_1 is at $\frac{n+1}{4}$, the median at $\frac{n+1}{2}$, Q_3 at $\frac{3(n+1)}{4}$. $IQR = Q_3 - Q_1$. To compare two sets, make **two** comparisons: one average and one spread, in context.

WHEN YOU SEE → YOU DO

“find the interquartile range”

→ order, then $Q_3 - Q_1$

11 values

→ $Q_1 = 3\text{rd}$, median = 6th, $Q_3 = 9\text{th}$

15 values

→ $Q_1 = 4\text{th}$, median = 8th, $Q_3 = 12\text{th}$

“compare”, “more consistent”

→ higher median = better on average;
smaller IQR = more consistent

WORKED MODEL • JANUARY 2022 P2H

Find the IQR of 8, 10, 12, 13, 5, 23, 21, 7, 5, 16, 14.

1 order: 5, 5, 7, **8**, 10, 12, 13, 14, **16**, 21, 23

$n = 11$

2 $Q_1 = 3\text{rd} = 7$, $Q_3 = 9\text{th} = 16$

positions

ANSWER $16 - 7 = 9 \text{ km}$

× **TRAP** Forgetting to put the list in order first.

IDEA 04

Practise: Quartiles, IQR and Comparing

My questions first, then selected practice, then real past papers.

Q53 DR ESLAM IQR

CH 6 • TOPIC 1 2 marks

Find the interquartile range of 3, 5, 6, 8, 9, 11, 12, 14, 15, 18, 20.

SELECTED PRACTICE

Q54

PRACTICE

Practice • Easy

CH 6 • TOPIC 1

2 marks

3 Twenty students took a Science test and a Maths test.

Both tests were marked out of 50

The table gives information about their results.

	Median	Interquartile range
Science	27	18
Maths	24.5	11

Use this information to compare the Science test results with the Maths test results.

Write down **two** comparisons.

(2 marks)

Q55**PRACTICE**

Practice • Easy

CH 6 • TOPIC 1 • 4 marks

- 8 15 students took an English test.
The same 15 students took a Maths test.
Both tests were marked out of 30.
For the English test results the median was 21
the interquartile range was 14

The Maths test results are shown below.

18	18	19	20	24	25	25	26	28	28	29	29	29	30	30
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Use the information above to compare the English test results with the Maths test results.
Write down **two** comparisons.

(4 marks)

Q56

PRACTICE

Practice • Medium

CH 6 • TOPIC 1 4 marks

- 4 (a) A bus company recorded the ages, in years, of the people on coach A and the people on coach B.

Here are the ages of the 23 people on coach A.

41	42	44	48	52	53	53	53	56	57	57	59
60	61	63	64	64	66	67	69	74	77	79	

Complete the table below to show information about the ages of the people on coach A.

Median	
Lower Quartile	
Upper Quartile	
Least age	41
Greatest age	79

(2 marks)

- (b) Here is some information about the ages of the people on coach B.

Median	70
Lower quartile	54
Upper quartile	73
Least age	42
Greatest age	85

Richard says that the people on coach A are younger than the people on coach B.

Is Richard correct?

You must give a reason for your answer.

(1 mark)

- (c) Richard says that the people on coach A vary more in age than the people on coach B.

Is Richard correct?

You must give a reason for your answer.

(1 mark)

Q57**PRACTICE**

Practice • Medium

CH 6 • TOPIC 1 4 marks

10 (a) Sandeep sat 11 tests in January 2020. Each test was marked out of 60.

Here are his test results.

45 41 35 44 38 47 47 39 37 43 42

Find the interquartile range of these test results. Show your working clearly.

(3 marks)

(b) Sandeep also sat some tests in May 2020.

Each test was marked out of 60.

The median of the May 2020 test results is 42.

The interquartile range of the May 2020 test results is 12.

In which month, January or May, were Sandeep's test results more consistent?

Give a reason for your answer.

(1 mark)

PAST PAPERS

Q58**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q14

CH 6 • TOPIC 1 3 marks

14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term.

Here are his marks.

18 5 7 12 11 18 15 16 17 13 14

Find the interquartile range of the marks.

.....
(Total for Question 14 is 3 marks)

Q59**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q12

CH 6 • TOPIC 1 2 marks

12 15 people were asked how long, in minutes, they had been waiting for a bus.

Here are the results.

2 3 3 4 5 6 6 8 9 10 11 13 14 15 18

Find the interquartile range of these times.

..... minutes
(Total for Question 12 is 2 marks)

Q60**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q14

CH 6 • TOPIC 1 2 marks

14 Here is the number of goals that Henri's team scored one summer in each water polo match.

5 8 9 11 13 13 14 15 16 17 20

Find the interquartile range of the numbers of goals.
Show your working clearly.

(Total for Question 14 is 2 marks)

Q61**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q15

CH 6 • TOPIC 1 3 marks

15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8 10 12 13 5 23 21 7 5 16 14

Find the interquartile range of his results.

..... km

(Total for Question 15 is 3 marks)

Q62**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q13

CH 6 • TOPIC 1**2 marks**

13 Here is the number of runs scored by a baseball team in each of its 15 games this season.

The number of runs have been arranged in order of size.

0 1 1 3 5 6 7 7 8 9 9 12 12 15 16

Work out the interquartile range of the number of runs.

.....

(Total for Question 13 is 2 marks)

Q63**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q14

CH 6 • TOPIC 1 4 marks

- 14 Akari played a computer game eleven times.
Here are her scores.

25 20 28 27 26 22 23 29 20 29 26

- (a) Find the interquartile range of her scores.

.....

Machi played the same computer game eleven times.

The interquartile range for Machi's scores was 9

- (b) Who had the more consistent scores, Akari or Machi?
Give a reason for your answer.

.....
.....

(Total for Question 14 is 4 marks)

Q64 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1H • Q10**CH 6 • TOPIC 1** **2 marks**

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

.....
(Total for Question 10 is 2 marks)

Q65 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 2H • Q13**CH 6 • TOPIC 1** **2 marks**

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

.....
(Total for Question 13 is 2 marks)

Q66**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q10

CH 6 • TOPIC 1**2 marks**

10 Here are the numbers of goals scored by a hockey team in its 11 games this season.

0 1 2 2 3 4 4 6 7 9 11

Work out the interquartile range of the numbers of goals.

.....

(Total for Question 10 is 2 marks)

Q67**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q13

CH 6 • TOPIC 1**2 marks**

13 Here are the numbers of eggs laid by some hens on each of 11 days in June.

9 10 12 15 17 18 19 19 20 25 26

Work out the interquartile range of the numbers of eggs.

.....

(Total for Question 13 is 2 marks)

Q68**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q16

CH 6 • TOPIC 1 2 marks

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

.....
(Total for Question 16 is 2 marks)

05 Charts: Pie, Bar and Time Series

A pie chart is shares of 360° : $\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360^\circ$. Bar charts and time series: read every value carefully, then use them.

WHEN YOU SEE → YOU DO

“draw the pie chart”

→ $360 \div \text{total} = \text{degrees per person}$

an angle is given

→ $\text{frequency} = \frac{\text{angle}}{360} \times \text{total}$

a time-series graph

→ read each point, then total or average

a bar chart and “the median”

→ running total of the bars

WORKED MODEL • NOVEMBER 2024 P2H

300 students; Biology has 126° on the pie chart. How many chose Biology?

1 $\frac{126}{360} \times 300$

share of the circle

ANSWER 105 students

IDEA 05

Practise: Charts: Pie, Bar and Time Series

My questions first, then selected practice, then real past papers.

Q69**DR ESLAM**

Pie chart

CH 6 • TOPIC 1

2 marks

In a pie chart of 72 people, 30 chose red. Work out the angle for red.

SELECTED PRACTICE

Q70

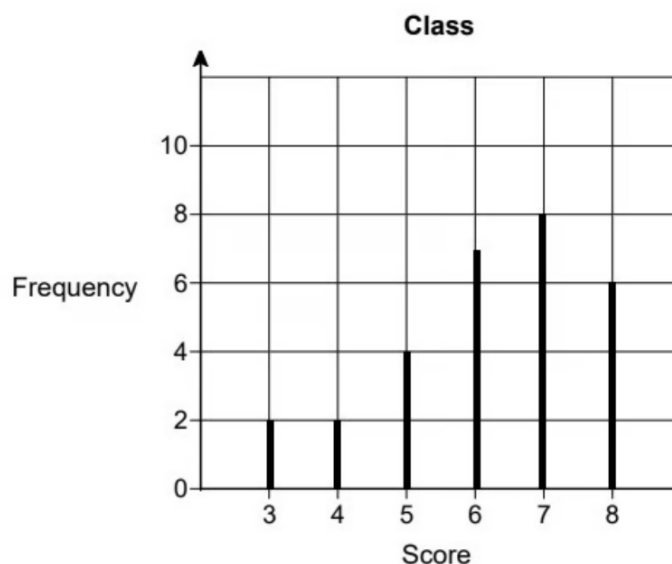
PRACTICE

Practice • Easy

CH 6 • TOPIC 1 2 marks

16 Students in a class took a spelling test.

The diagram shows information about the scores.



Lucy is one of the 29 students in the class.

Her score was the same as the **median** score for her class.

Work out her score.

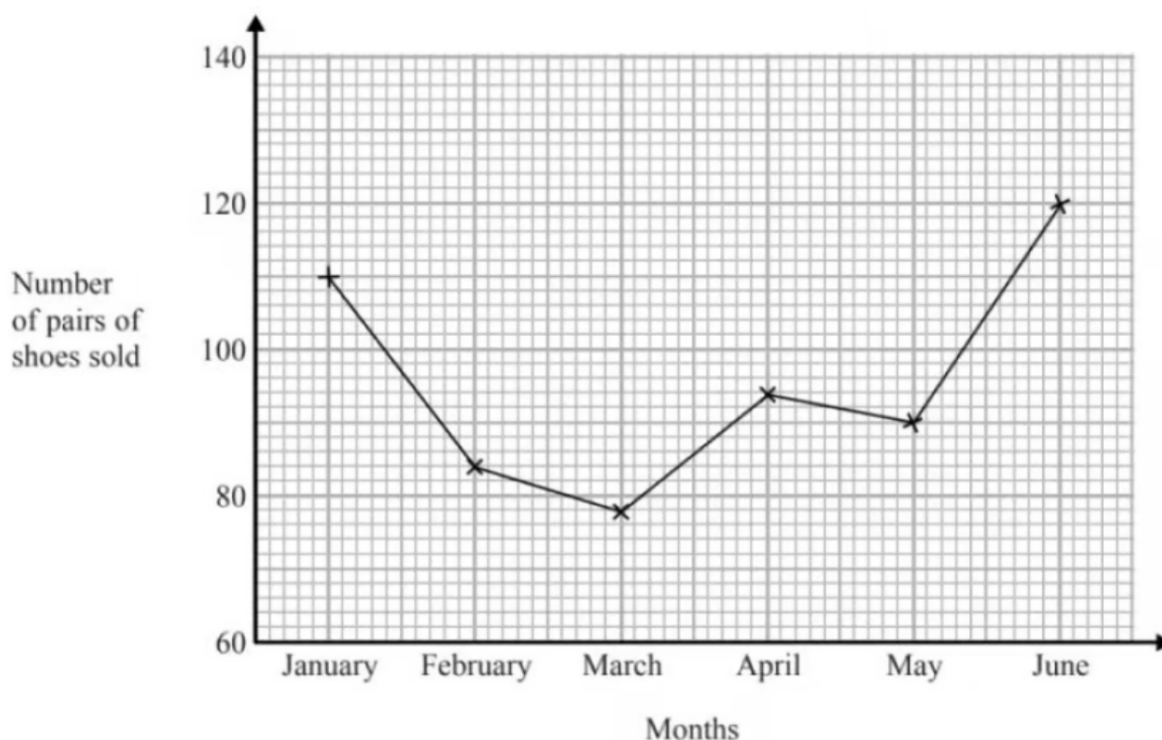
(2 marks)

Q71**PRACTICE**

Practice • Medium

CH 6 • TOPIC 1 **3 marks**

- 3 The time-series graph gives some information about the number of pairs of shoes sold in a shoe shop in the first six months of 2014.



The sales target for the first six months of 2014 was to sell a mean of 96 pairs of shoes per month.

Did the shoe shop meet this sales target?

You must show how you get your answer.

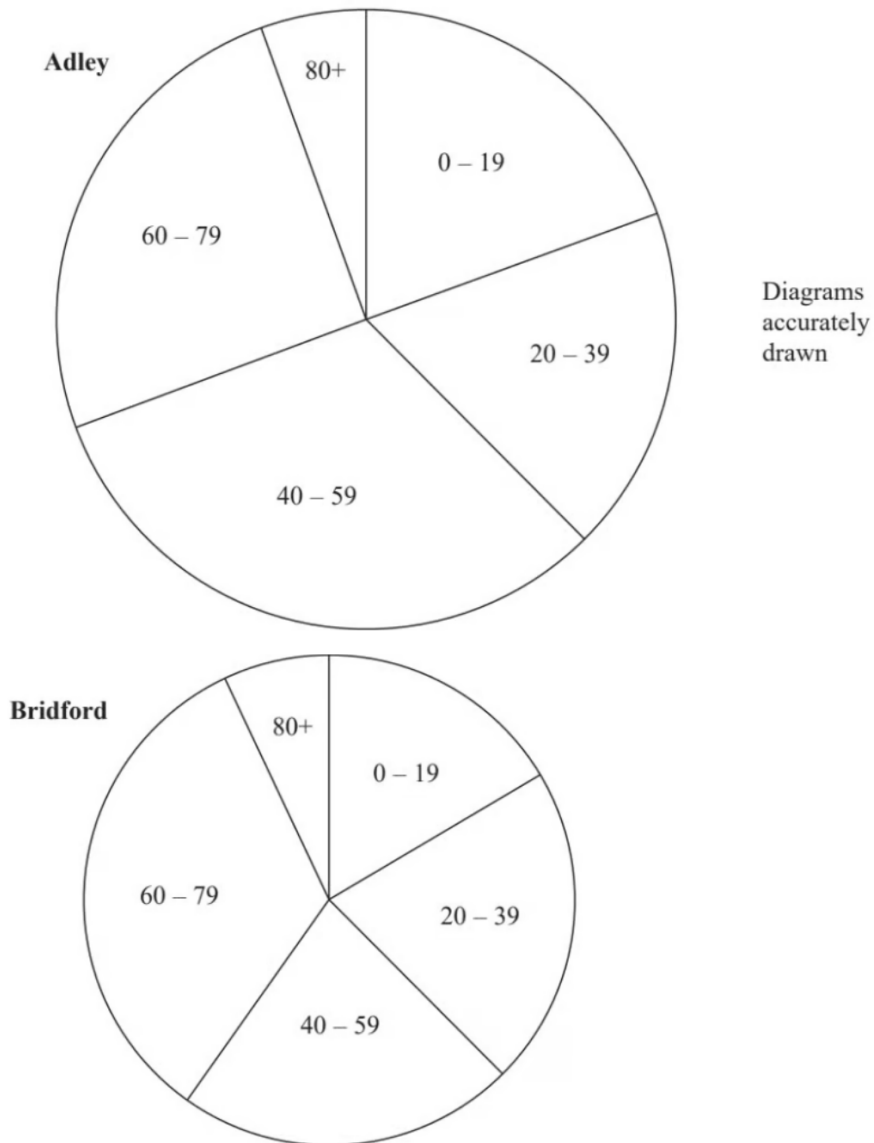
(3 marks)

Q72**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1 **3 marks**

- 15** The pie charts give information about the ages, in years, of people living in two towns, Adley and Bridford.



The ratio of the number of people living in Adley to the number of people living in Bridford is given by the ratio of the areas of the pie charts.

What proportion of the total number of people living in these two towns live in Adley and are aged 0 – 19?

Give your answer correct to 3 significant figures.

(3 marks)

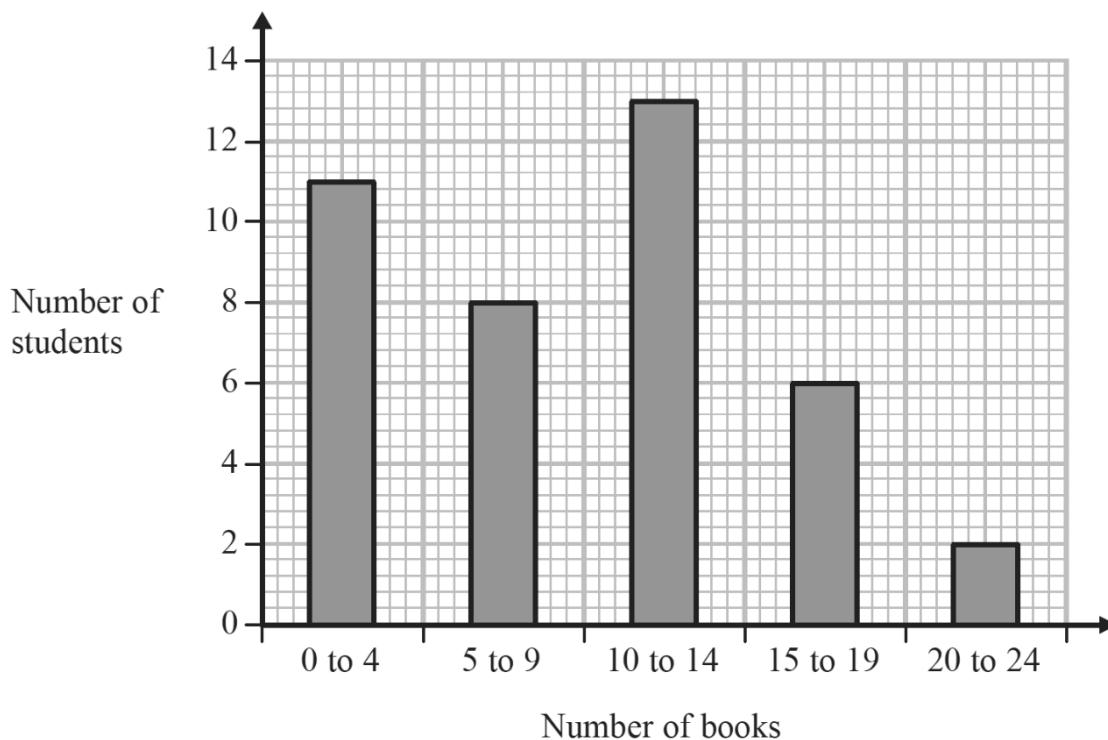
Q73**PRACTICE**

Practice • Hard

CH 6 • TOPIC 1 3 marks

17 (a) Fran asks each of 40 students how many books they bought last year.

The chart below shows information about the number of books bought by each of the 40 students.



Work out the percentage of these students who bought 20 or more books.

.....%

(2 marks)

(b) Show that an estimate for the mean number of books bought is 9.5
You must show all your working.

(1 mark)

PAST PAPERS

Q74

PAST PAPER

Edexcel 4MA1 • November 2024 • Paper 2H • Q10

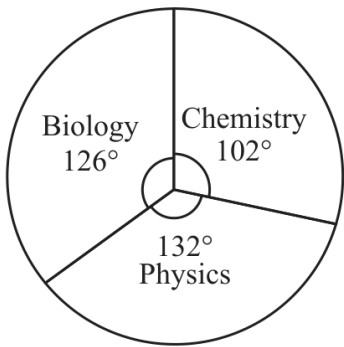
CH 6 • TOPIC 1

5 marks

PRACTISE • 6.1

10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.
The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10
There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

(Total for Question 10 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Averages | Order first; boxes; fill what each fact forces. |
| 02 | Mean problems | Mean $\times$ number = total; work with totals. |
| 03 | Grouped tables | Midpoints; $\sum fx \div \sum f$; modal class. |
| 04 | IQR | $Q_3 - Q_1$; compare an average and a spread. |
| 05 | Charts | Angle = $\frac{f}{\text{total}} \times 360^\circ$. |

BOTTOM LINE Means $\rightarrow$ totals $\cdot$ groups $\rightarrow$ midpoints $\cdot$ compare $\rightarrow$ average and spread, in context.

Answers

Statistics • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • AVERAGES AND RANGE FROM A LIST

- Q1** Dr Eslam • Averages Mean $48 \div 8 = 6$; median 6; mode 9; range 7
- Q2** Dr Eslam • Missing numbers 5, 5, 7, 11
- Q3** Practice • Easy Highest $39 + 57 = 96$, lowest 33: range 63
- Q4** Practice • Easy Total 60; the other two add to 23 and are 2 and 21
- Q5** Practice • Easy 69, 76, 76, 79
- Q6** Practice • Medium $7 + a + b = 13$ and $c + 8 = 13$: e.g. $a = 2$, $b = 4$, $c = 5$
- Q7** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q4 $x = 8$, $y = 18.5$
- Q8** Edexcel 4MA1 • January 2021 • Paper 1HR • Q2 $a = 7$, $b = 10$, $c = 12$
- Q9** Edexcel 4MA1 • January 2021 • Paper 2H • Q2 3, 6, 7, 8, 8
- Q10** Edexcel 4MA1 • January 2022 • Paper 1HR • Q5 64, 73, 80
- Q11** Edexcel 4MA1 • January 2022 • Paper 1H • Q5 5, 8, 8, 20, 21, 24 and 33
- Q12** Edexcel 4MA1 • June 2022 • Paper 2HR • Q3 Candela = 162 cm, Diana = 154 cm
- Q13** Edexcel 4MA1 • June 2022 • Paper 2H • Q1 $a = 5$, $b = 6$, $c = 7$, $d = 9$
- Q14** Edexcel 4MA1 • June 2024 • Paper 2HR • Q3 $x = 11$, $y = 14$
- Q15** Edexcel 4MA1 • June 2024 • Paper 2H • Q1 $h = 5$, $j = 12$, $k = 18$
- Q16** Edexcel 4MA1 • June 2025 • Paper 2H • Q1 4, 6, 9
- Q17** Edexcel 4MA1 • November 2025 $e = 2$, $f = 6$, $g = 10$

IDEA 02 • MEAN PROBLEMS: WORK WITH TOTALS

		$90 - 70 = 20$
Q18	Dr Eslam • Totals	
		$(19 \times 82 + 93) \div 20 = 82.55 \text{ kg}$
Q19	Practice • Easy	
		$11 \times 33 - 10 \times 35 = 13 \text{ points}$
Q20	Practice • Hard	
		$11 \times 33 - 10 \times 35 = 363 - 350 = 13$: Jordan is right
Q21	Practice • Hard	
		Mean A = 141, mean B = 376; $5 \times 376 - 1453$: $x = 427$
Q22	Practice • Hard	
		29 kg
Q23	Edexcel 4MA1 • January 2020 • Paper 1HR • Q8	
		416 grams
Q24	Edexcel 4MA1 • January 2021 • Paper 2H • Q7	
		At least 92
Q25	Edexcel 4MA1 • May 2021 • Paper 1H • Q10	
		185 kg
Q26	Edexcel 4MA1 • January 2022 • Paper 2HR • Q8	
		$x = 17$
Q27	Edexcel 4MA1 • June 2022 • Paper 1HR • Q3	
		1.9 kg
Q28	Edexcel 4MA1 • June 2022 • Paper 2H • Q12	
		237 km
Q29	Edexcel 4MA1 • June 2023 • Paper 2HR • Q10	
		12
Q30	Edexcel 4MA1 • June 2023 • Paper 2H • Q10	
		$k = 6$
Q31	Edexcel 4MA1 • November 2023 • Paper 2H • Q8	
		12403
Q32	Edexcel 4MA1 • June 2024 • Paper 2H • Q12	
		$x = 21$
Q33	Edexcel 4MA1 • May 2024 • Paper 1HR • Q1	
		130 g
Q34	Dr Eslam Ahmed • eliteigcse.com • WhatsApp 011 2000 9622 Edexcel 4MA1 • November 2024 • Paper 2H • Q8	

IDEA 03 • GROUPED FREQUENCY TABLES

Q36	Dr Eslam • Grouped mean	$(5 \cdot 4 + 15 \cdot 10 + 25 \cdot 6) \div 20 = 320 \div 20 = 16$
Q37	Practice • Easy	$0 < p \leq 1$
Q38	Practice • Medium	$9860 \div 200 = 49.3$ minutes
Q39	Practice • Hard	$4650 \div 40 = 116.25$ minutes
Q40	Practice • Hard	(a) $1684 \div 50 = 33.68$ inches (b) Small is $\frac{24}{50} = 48\%$, not 75%: the manager is correct
Q41	Edexcel 4MA1 • January 2020 • Paper 1H • Q4	(a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$
Q42	Edexcel 4MA1 • January 2020 • Paper 2HR • Q2	£2920
Q43	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q4	Modal class $30 < t \leq 40$, estimated mean 28.2 minutes
Q44	Edexcel 4MA1 • January 2021 • Paper 1H • Q2	(a) $70 < s \leq 80$, (b) 66.4 km/h
Q45	Edexcel 4MA1 • January 2021 • Paper 2HR • Q10	$x = 10$
Q46	Edexcel 4MA1 • November 2021 • Paper 2H • Q3	$\frac{31}{70}$, estimated mean 21.3 euros
Q47	Edexcel 4MA1 • January 2022 • Paper 2H • Q9	$x = 21$
Q48	Edexcel 4MA1 • May 2023 • Paper 1H • Q2	2160 minutes
Q49	Edexcel 4MA1 • November 2023 • Paper 2H • Q1	Modal class $25 < m \leq 30$, estimated total 945 minutes
Q50	Edexcel 4MA1 • November 2024 • Paper 1H • Q1	Modal class $10 < p \leq 15$, estimated mean 19.5 dollars
Q51	Edexcel 4MA1 • June 2025 • Paper 1H • Q1	Modal class $5 < d \leq 10$, estimated mean 9.2 km
Q52	Edexcel 4MA1 • November 2025 • Paper 1H • Q1	11 km

IDEA 04 • QUARTILES, IQR AND COMPARING

Q53	Dr Eslam • IQR	$Q_1 = 6$ (3rd), $Q_3 = 15$ (9th): IQR = 9
Q54	Practice • Easy	Science median higher: better on average. Maths IQR smaller: more consistent
Q55	Practice • Easy	Maths: median 26, IQR $29 - 20 = 9$. Maths better on average ($26 > 21$) and more consistent ($9 < 14$)
Q56	Practice • Medium	(a) median 59, LQ 53, UQ 66 (b) Yes: median $59 < 70$ (c) No: IQR $13 < 19$ (range $38 < 43$), so coach A varies less
Q57	Practice • Medium	(a) $45 - 38 = 7$ (b) January: smaller IQR ($7 < 12$)
Q58	Edexcel 4MA1 • January 2020 • Paper 2HR • Q14	6
Q59	Edexcel 4MA1 • January 2021 • Paper 2H • Q12	9 minutes
Q60	Edexcel 4MA1 • May 2021 • Paper 1H • Q14	7
Q61	Edexcel 4MA1 • January 2022 • Paper 2H • Q15	9 km
Q62	Edexcel 4MA1 • June 2022 • Paper 1HR • Q13	9
Q63	Edexcel 4MA1 • June 2023 • Paper 2HR • Q14	Akari's IQR is 6, and Akari is more consistent
Q64	Edexcel 4MA1 • May 2023 • Paper 1H • Q10	10
Q65	Edexcel 4MA1 • November 2023 • Paper 2H • Q13	9
Q66	Edexcel 4MA1 • May 2024 • Paper 1HR • Q10	5
Q67	Edexcel 4MA1 • June 2025 • Paper 1HR • Q13	8
Q68	Dr Eslam Ahmed • eliteigcse.com • WhatsApp 011 2000 9622 Edexcel 4MA1 • November 2025 • Paper 1H • Q16	21

IDEA 05 • CHARTS: PIE, BAR AND TIME SERIES

- Q69** Dr Eslam • Pie chart $\frac{30}{72} \times 360 = 150^\circ$
- Q70** Practice • Easy 29 students: 15th score = 6
- Q71** Practice • Medium $(110 + 84 + 78 + 94 + 90 + 120) \div 6 = 576 \div 6 = 96$:
yes, target met
- Q72** Practice • Hard Areas from the radii, then $\times$ the 0–19 angle
fraction: about 11.9%
- Q73** Practice • Hard (a) $\frac{2}{40} = 5\%$ (b) $(22 + 56 + 156 + 102 + 44) \div 40 =$
 $380 \div 40 = 9.5$
- Q74** Edexcel 4MA1 • November 2024
• Paper 2H • Q10 36 more students